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Are COVID-19 models blind to the social determinants of health? A systematic review protocol

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Are COVID-19 models blind to the social determinants of health? A systematic review protocol

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ABSTRACT

Introduction: Infectious disease models are important tools to inform public health policy decisions. These models are primarily based on an average population approach and often ignore the role of social determinants in predicting the course of pandemic and the impact of policy interventions. Ignoring social determinants in models may cause or exacerbate inequalities. This limitation has not been previously explored in the context of the current pandemic, where COVID-19 has been found to disproportionately affect marginalized racial, ethnic, and socioeconomic groups. Therefore, our primary goal is to identify the extent to which COVID-19 models incorporate the social determinants of health in predicting outcomes of the pandemic.

Methods and analysis: We will search MEDLINE, EMBASE, Cochrane Library, and Web of Science databases from 2019 to present. We will include all infectious disease modelling studies that meet all of the following criteria: a) focused on human spread of SARS-CoV-2; b) modelling studies; c) interventional or non-interventional studies; and d) focused on one of the following outcomes: COVID-19 related outcomes (e.g. cases, deaths, etc.), non-COVID-19 related outcomes (i.e. impacts of the pandemic or control policies on other health conditions or health services), or impact of the pandemic or control policies on economic outcomes. We will report the percentage of included models that considered social factors, indicate which social factors were considered, and describe how social factors were incorporated into the conceptualization and implementation of the infectious disease models. The extracted data will also be used to create a narrative synthesis of the results.

Ethics and dissemination: Ethical approval is not required as only secondary data will be collected. The results of this systematic review will be disseminated through peer-reviewed publication and conference proceedings.

Systematic review registration: PROSPERO CRD42020207706

Keywords: COVID-19, epidemiology, public health, social medicine

ARTICLE SUMMARY

Strengths and limitations of this study

- The PRISMA-P (Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols) will be followed.
- A systematic approach to evaluating modellers' incorporation of social factors into COVID-19 models will identify practical approaches to better quantify social inequities of the pandemic and predict the impact of public policies on marginalized populations.
- Included studies of COVID-19 models may not be amenable to quantitative synthesis due to heterogeneity of the literature.
- Given the timeframe of the search strategy and the rapid evolution of COVID-19 modelling literature, the results may not include studies evaluating certain prevention strategies, such as vaccination.

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INTRODUCTION

In the early months of the pandemic, COVID-19 was described by New York Governor Andrew Cuomo as the 'great equalizer'[1] among all individuals as the new coronavirus swept through society without bias. A shared lack of immunity meant that regardless of prestige, wealth, fame, or age, we were all equally at-risk.[1] However, epidemiological researchers quickly demonstrated that this was, and remains to be, far from the truth. The pandemic has put existing health inequalities under the microscope and highlighted that COVID-19 has disproportionately affected marginalized racial, ethnic, and socioeconomic groups through its unequal health burden and its disparity of economic losses.[2, 3] A recent investigation found that among 158 counties in large U.S. metropolitan areas, those with predominantly non-White populations had infection rates approximately eight times higher, and a fatality rate more than nine times higher than counties with a mostly White population.[4] Similarly, people living in the poorest neighbourhoods of England and Wales have been found to be twice as likely to die from COVID-19 as those in more affluent areas.[5] Examples of similar trends can be found throughout history. Perhaps most notably in terms of size and scale, data collected from the 1918 flu pandemic demonstrated that non-White minorities had higher all-cause mortality and influenza-related mortality rates when compared to their White counterparts.[6] These disparities are a symptom of deeper societal and health system inequities, including disproportionate exposure through high risk jobs, prevalence of comorbidities, and inequitable access to testing and treatment.

Over the past century, infectious disease models have played a fundamental role in helping public health policy makers implement infection control interventions, as well as shape the theories utilized in disease mitigation.[7] Infectious disease models have been described as,

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"a scientific approach to formulating an explanation for an observed phenomenon and then testing this formulation to project the outcome of various experiments under pertinent conditions.[8(p1)]" Models provide a framework for experts to study a number of important factors related to the SARS-CoV-2 pandemic, including the following:

1) Viral transmission dynamics – models use observed data such as the number of confirmed cases, to estimate difficult to observe transmission dynamics, such as the incubation period and percentage of asymptomatic spread in the population[9, 10]
2) Anticipate the future course of an outbreak – models forecast future incidence, prevalence, morbidity and mortality, based on input parameters[11, 12]
3) Guide public health planning and infectious disease control – models help researchers simulate real-world possibilities and thus appraise the potential impact of disease control strategies such as mask wearing, school closures, testing and contract-tracing on the population[13, 14]
4) Models can estimate a range of outcomes, those directly related to COVID-19 or secondary effects of the disease and control measures, on population health, health service utilization and economic outcomes such as cost and quality of life.[15]

Infectious disease models can only be fit for purpose if the models are appropriately conceptualized and model parameters based on the best available evidence.[16, 17] Model conceptualization requires decisions about the model structure, including which health states to model, division of the population into sub-groups and selection of characteristics to define the subgroups.[16] Thus, models can incorporate the social determinants of health,[18] such as income, education, employment, and ethnicity, however structuring models to reflect social determinants increases complexity and sometimes, due to limited availability of data, raises

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questions about structural uncertainty.[19, 20] Complexity and uncertainty may lead modellers to simplify and favour average population models foregoing appraisal of different types of individuals or important subgroups.[19,20,21] Current evidence highlights the importance of social factors in predicting the course of the pandemic and in analyses of policy options,[2-6] however it is unclear to what extent current infectious disease models are based on an average population approach and ignore social factors. To the best of our knowledge, this limitation has yet to be explored in the context of the COVID-19 pandemic. Therefore, the present systematic review will aim to answer the following research question: "To what extent do COVID-19 models incorporate the social determinants of health?" We hypothesize that by providing a complete synthesis of all human COVID-19 modelling (predictive, forecasting) studies to date, the findings will highlight the paucity of models that incorporate social factors into the predicted outcomes of the pandemic. In those modelling studies that account for social factors, this review will identify how they were incorporated into the conceptualization of the infectious disease models and provide a blueprint for other modellers to incorporate social factors in the future. We hope that this review will lead to infectious disease models that explicitly acknowledge the role of social determinants in making better predictions as well as mitigating inequalities, and therefore result in better infectious disease outcomes.

METHODS

Protocol and registration

This systematic review will be conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines[22] for the identification, screening, eligibility, and inclusion of retrieved research studies. The current protocol is

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registered in the International Prospective Register of Systematic Reviews (PROSPERO)[23] (Registration number CRD42020207706).

Search strategy and eligibility criteria

MEDLINE, EMBASE, Cochrane Library, and Web of Science databases will be searched for published studies. Preprints will be identified in MedRxiv, OSF Preprints, SSRN, and the Society for Medical Decision Making (SMDM) COVID-19 Decision Model Repository. Additionally, unpublished national-level models and repositories will be identified through national public health websites and the SMDM network. The present systematic review is designed to have broad eligibility criteria to identify all COVID-19 modelling studies from 2019 to present. Studies will be included if they: 1) have a human SARS-CoV-2 population, 2) are a modelling study (e.g. mechanistic, mathematical, network, simulation, auto-regressive, Markov, or other model), and 3) investigate one of the following outcomes: COVID-19 disease-related outcomes (e.g. spread, hospitalizations, deaths, recoveries, etc.), non-COVID-19 disease-related outcomes (e.g. impact on other health conditions such as mental health), impacts on health services (e.g. bed occupancy, ventilators, surgical delays), impact of policies (e.g. lock-downs, social distancing) or interventions (vaccines, treatments, etc.) on COVID-19 outcomes or its societal impact. Eligible studies will then be assessed for components of social determinants. The following studies will be excluded: 1) within-host biological studies, 2) non-human SARS-CoV-2 studies, 3) phylogenic/genetic studies, 4) environmental/meteorological studies without health impact analysis on humans, 5) epidemiological/statistical analyses without a model, and 6) economic analyses without a model. An example search strategy can be found in Table 1.

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Table 1. MEDLINE search strategy (August 4, 2020)

1. coronavirus/ or betacoronavirus/ or coronavirus infections/ (20740)
2. ("coronavirus*" or "coron?virinae*" or "novel coronavirus*" or "novel corona virus" or "Severe Acute Respiratory Syndrome Coronavirus 2" or "coronavirus disease 2019" or "coronavirus pandemic?" or "coronavirus epidemic?" or "coronavirus outbreak?" or "corona virus pandemic?" or "corona virus epidemic?" or "corona virus outbreak?" or "corona virus disease 2019" or "new coronavirus" or "new corona virus*" or "new coronaviruses" or "novel coronaviruses" or "2019 ncov" or "nCov 2019" or "SARS Coronavirus 2" or "2019-nCoV" or "2019nCoV" or "2019-CoV" or "nCoV2019" or "nCoV-2019" or "COVID-19" or "COVID19" or "CORVID-19" or "CORVID19' or WN-CoV or WNCoV'" or "HCoV-19" or "HCoV19" or "CoV " or "2019 novel*" or "Ncov" or "nCov" or "n-cov" or "SARSCoV-2" or "SARSCoV-2" or "SARSCoV2" or "SARS-CoV2" or "SARS-COV-2" or "SARSCov19" or "SARS-Cov19" or "SARSCov-19" or "SARS-Cov-19" or "SARSr-cov" or "Ncovor" or "Ncorona*" or "Ncorono*" or "NcovWuhan*" or "NcovHubei*" or "NcovChina*" or "NcovChinese*" or "Wuhan virus*" or "novel CoV" or "CoV 2" or "CoV2" or "betacoron?vir*").mp. (54405)
3. ((corona* or corono*) adj1 (virus* or viral* or virinae*)).mp. (1032)
4. (((respiratory* adj2 (acute* or symptom* or disease* or illness* or infect* or condition*)) or "sea-food market*" or "seafood market*" or "food market*" or "foodmarket*" or "wet market*" or "wet-market*" or "wetmarket*") adj10 (Wuhan* or Hubei* or China* or Chinese* or Huanan*)).mp. (1479)
5. ((outbreak* or wildlife* or wild-life or pandemic* or epidemic* or coronavirus or corona virus) adj3 (Wuhan* or Hubei* or China* or Chinese* or Huanan*)).mp. (2228)
6. (anti-flu* or anti-influenza* or antiflu* or antinfluenza*).mp. (3379)
7. or/1-6 (60113) - **(All Coronavirus / COVID-19 Terms)**
8. Models, Theoretical/ (150928)
9. Models, Biological/ (338438)
10. Stochastic Processes/ or stochastic.mp. (44593)
11. (forecast* or model* or simulat*).af. (4238913)
12. or/8-11 (4250384) - **(All Model Terms)**
13. 7 and 12 (6971) - **(All Coronavirus / COVID-19 Terms) AND (All Model Terms)**
14. transmi*.ti,kf. (110754)
15. 7 and 14 (1788) – **(Result Set to capture a non-COVID-19 paper)**
16. 13 or 15 (8442)
17. ("32691014" or "32689711" or "32171948" or "32687538" or "32685697" or "32685143" or "32673577" or "32660125" or "32546824" or "32372755" or "32246905" or "32325039" or "32444481" or "32269020" or "32245814" or "32632012" or "22999128" or "22990082" or "31422772" or "28577700" or "24034486" or "32691015" or "32690354" or "32680824" or "32298421" or "22999128" or "22990082" or "22999132" or "31422772" or "28577700" or "24034486").ui. (26)
18. 13 and 17 (20)
19. limit 16 to yr="2019 -Current" (4850) - **(Final Result Set)**

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Review process

After executing the search strategy, articles will be uploaded to Covidence Software (v204501fd4fa9) and duplicates will be removed. Titles and/or abstracts of the articles will be screened independently by two review members to identify studies that potentially meet the inclusion criteria outlined above. Conflicts will be resolved by a third independent reviewer. The full texts of these potentially eligible studies will then be retrieved and independently assessed for eligibility by two review team members. Any disagreements between them over the eligibility of particular studies will be resolved by a third reviewer. Data will then be extracted from the studies that incorporated social factors. A flow chart for this systematic review can be found in Figure 1.

Data extraction and quality assessment

Data will be extracted into Covidence by one reviewer and confirmed by a second reviewer. Disagreements will be resolved with a discussion between the two reviewers and if necessary, the final decision will be made by a third reviewer. If data related to the outcomes are missing, this will be reported. Authors will not be contacted to request missing data. The following data will be extracted from included studies: author, publication year, geography (country, province/state, city/town, region, low/middle/high income country), setting (Ex. general population, long-term care home, hospital, workplace etc.), population size, goals (predicting the pandemic, impact on other health conditions, impact on health services, impact of policy or interventions, economic analyses), type of model (mathematical, mechanistic, etc.), social and economic factors, and approach to incorporating social and economic factors. An example extraction table containing the characteristics of included studies in the systematic review can be found in Table 2.

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Table 2. Example Extraction Table - Characteristics of Included Studies

Author(s)	Month/ Year	Geography	Setting	(N)	Goal(s)	Model Type	Social Factors	Approach to Social Factors
Smith et al.	March 2020	Milan, Italy High income	Hospital	400	Predicting the Pandemic Case Fatality	Compartmental	Income Employment	TBD
Jones et al.	April 2020	Wuhan, China High income	General	2000	Impact of Policy or Intervention Lock-down	Network	Education	TBD
Harrison et al.	May 2020	Seattle, U.S. High income	Long-term care	300	Impact on other health conditions Mental Health	Mechanistic	Physical Environment	TBD

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There is currently no standardized tool for assessing the quality of infectious disease modelling studies. We plan to adapt the principles of best practices for infectious disease modelling, laid out by Pitman and colleagues into a tool for quality assessment of infectious disease models.[24]

Evidence synthesis

The extracted data will be used to create a narrative synthesis of the results. Based on our broad inclusion criteria, it is likely that the study designs and data will be too heterogeneous for meta-analyses to be appropriate. We expect to report on the number (percent) of identified infectious disease models that incorporated social factors. We will report on the social factors considered, and indicate the number (percent) of included models that considered each social factor. The narrative synthesis will describe how social factors were incorporated into the conceptualization of the infectious disease models, estimate policy impact across social groups (if relevant), and describe which model types (e.g. compartmental, network models) were used when social factors were considered.

DISCUSSION

To the best of our knowledge, this will be the first systematic review that aims to identify the frequency and approach used in COVID-19 models to incorporate the social determinants of health to predict outcomes of the pandemic. Despite previous research and adequate knowledge of the disproportionate health burden experienced by marginalized groups, we hypothesize that the majority of COVID-19 models continue to ignore these important social factors. Without taking into account the influences of unequal levels of exposure, vulnerability, and inequitable access to health services dictated by social factors,[25] COVID-19 models may be providing an unclear illustration of the impact of COVID-19 and disease control efforts. When health system

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decision-makers utilize average population-based models to implement large-scale health policies, they may be unintentionally exacerbating existing health inequities and leaving these marginalized groups further behind. Furthermore, failure to incorporate social factors may lead to missed opportunities to appraise targeted interventions to overcome inequities in the impact of COVID-19. Indeed, interventions targeted to socially vulnerable groups may prove more efficient for controlling the pandemic than interventions at the population level. Although our primary goal is to solidify this fact into scientific literature, our secondary goal with this review is to identify modellers that have incorporated social factors into the conceptualization of their COVID-19 models, and detail the approach they used to do so. It is plausible that the lack of social factors incorporated into COVID-19 models is not a result of neglect, but due to time constraints and limited computational power to handle increased model complexity. The modelling community is highly specialized and their skill sets are coveted by a range of policy decision-makers. Thus, modellers can have significant influence over policy decision-making processes. If modellers increase the awareness of the need to incorporate social factors into COVID-19 models, constraints on incorporating social factors into models can potentially be overcome through increased investments in computer infrastructure and strategic collaboration amongst modelling groups. It is our hope that this review will help provide a preliminary roadmap for modellers to incorporate social factors into future COVID-19 models, and subsequently result in better health outcomes for the entire population.

The proposed review has some limitations. First, while we will make an effort to include pre-print studies and identify models used by public health agencies, it is challenging to be comprehensive in the search for global grey literature. This is due to the large number of COVID-19 models described on government web-sites that have been used to inform policy at

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the global, federal, state/provincial or regional level. However, we do not expect that models in the grey literature will be more complex than published models and thus, grey literature models will be no more likely to incorporate social factors than published models. Second, COVID-19 modelling literature is rapidly evolving. Given the timeframe of the search strategy, we will likely not find many studies evaluating vaccination strategies. However, the findings on incorporating social factors into infectious disease models will still be relevant to modellers working on vaccine policy appraisal.

The strengths of the proposed review are that a systematic approach to evaluating modellers' incorporation of social factors into COVID-19 models can identify practical approaches to making better predictions and reduce inequities, quantify gaps in the literature, motivate enhancements in modelling best practices, and ultimately improve health for socially and economically vulnerable groups related to COVID-19 and other health conditions.

ETHICS AND DISSEMINATION

Ethical approval is not required as only secondary data will be collected. The results of this systematic review will be disseminated through peer-reviewed publication and conference proceedings.

PATIENT AND PUBLIC INVOLVEMENT

There were no time or funds allocated to patient and public involvement due to safety concerns in the context of the current COVID-19 pandemic. However, this systematic review asks an important research question relevant to the health and protection of vulnerable groups.

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REFERENCES

1. Mein SA. COVID-19 and Health Disparities: the Reality of "the Great Equalizer". *J Gen Intern Med* 2020;35(8):2439-40. doi: 10.1007/s11606-020-05880-5

2. Ali S, Asaria M, Stranges S. COVID-19 and inequality: are we all in this together?. *Can J Public Health* 2020;111:415-416. doi: 10.17269/s41997-020-00351-0

3. Williams DR, Cooper LA. COVID-19 and health equity—a new kind of "herd immunity". *JAMA* 2020;323(24):2478-2480. doi: 10.1001/jama.2020.8051

4. Adhikari S, Pantaleo NP, Feldman JM, et al. Assessment of community-level disparities in coronavirus disease 2019 (COVID-19) infections and deaths in large US metropolitan areas. *JAMA Network Open* 2020;3(7):e2016938-e2016938. doi: 10.1001/jamanetworkopen.2020.16938

5. Caul S. Deaths involving COVID-19 by local area and socioeconomic deprivation: deaths occurring between 1 March and 31 July 2020. Statistical Bulletin. 2020.

6. Hutchins SS, Fiscella K, Levine RS, et al. Ompad, D. C., & McDonald, M. Protection of racial/ethnic minority populations during an influenza pandemic. *Am J Public Health* 2009;99(S2):S261-S270. doi: 10.2105/AJPH.2009.161505

7. Lessler J, Cummings DA. Mechanistic models of infectious disease and their impact on public health. *Am J Epidemiol* 2016;183(5):415-422. doi: 10.1093/aje/kww021

8. Star L, Moghadas SM. The role of mathematical modelling in public health planning and decision making. Purple Paper, National Collaborative Center for Infectious Diseases. 2010.

9. Zhou L, Liu JM, Dong XP, McGoogan JM, et al. & Wu, Z. Y. COVID-19 seeding time and doubling time model: an early epidemic risk assessment tool. *Infect Dis Poverty* 2020; 9(76):1-9. doi: 10.1186/s40249-020-00685-4

10. Tuite AR, Fisman DN. Reporting, epidemic growth, and reproduction numbers for the 2019 novel coronavirus (2019-nCoV) epidemic. *Ann Intern Med* 2020;172(8):567-568. doi: 10.7326/M20-0358

11. Wang Y, Xu C, Yao S, et al. Forecasting the epidemiological trends of COVID-19 prevalence and mortality using the advanced α-Sutte Indicator. *Epidemiol Infect* 2020;148:e236. doi: 10.1017/S095026882000237X

12. Ogden NH, Fazil A, Arino J, et al. Artificial intelligence in public health: Modelling scenarios of the epidemic of COVID-19 in Canada. *Can Commun Dis Rep* 2020;46(8):198-204. doi: 10.14745/ccdr.v46i06a08

13. Fisman DN, Greer AL, Tuite AR. Bidirectional impact of imperfect mask use on reproduction number of COVID-19: A next generation matrix approach. *Infect Dis Model* 2020;5:405-408. doi: 10.1016/j.idm.2020.06.004

14. Tuite AR, Fisman DN, Greer AL. Mathematical modelling of COVID-19 transmission and mitigation strategies in the population of Ontario, Canada. *CMAJ* 2020;192(19): E497-E505. doi: 10.1503/cmaj.200476

15. Wells CR, Fitzpatrick MC, Sah P, et al. Projecting the demand for ventilators at the peak of the COVID-19 outbreak in the USA. *Lancet Infect Dis* 2020;20(10):1123-1125. doi: 10.1016/S1473-3099(20)30315-7

16. Roberts M, Russell LB, Paltiel AD, et al. Conceptualizing a model: a report of the ISPOR-SMDM modeling good research practices task force–2. *Med Decis Making* 2012;32(5): 678-689. doi: 10.1177/0272989X12454941

17. Briggs AH, Weinstein MC, Fenwick EA, et al. Model parameter estimation and uncertainty analysis: a report of the ISPOR-SMDM Modeling Good Research Practices Task Force Working Group–6. *Med Decis Making*, 2012;32(5):722-732. doi: 10.1177/0272989X12458348

18. Marmot M. Social determinants of health inequalities. *Lancet Public Health* 2005;365(9464): 1099-1104. doi: 10.1016/S0140-6736(05)71146-6

19. Mauskopf J. Multivariable and structural uncertainty analyses for cost-effectiveness estimates: back to the future. *Value Health* 2019;*22*(5): 570-574. doi: 10.1016/j.jval.2018.11.013

20. Jain R, Grabner M, Onukwugha E. Sensitivity analysis in cost-effectiveness studies. *Pharmacoeconomics* 2011;29(4):297-314. doi: 10.2165/11584630-000000000-00000

21. Sculpher M. Subgroups and heterogeneity in cost-effectiveness analysis. *Pharmacoeconomics* 2008;26(9):799-806. doi: 10.2165/00019053-200826090-00009

22. Mohers D, Shamseer L, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Syst Rev* 2015;*4*(1): 1. doi: 10.1186/2046-4053-4-1

23. Booth A, Clarke M, Dooley G, et al. The nuts and bolts of PROSPERO: an international prospective register of systematic reviews. *Syst Rev* 2012;1(2): 1-9. doi: 10.1186/2046-4053-1-2

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24. Pitman R, Fisman D, Zaric GS, et al. Dynamic transmission modeling: a report of the ISPOR-SMDM Modeling Good Research Practices Task Force Working Group–5. *Med Decis Making* 2012;32(5):712-721. doi: 10.1177/0272989X12454578

25. Moloughney BW. Social Determinants of Health: What can public health do to address inequities in infectious disease?. *Can Commun Dis Rep* 2016;42(Suppl 1): S1-14-S1-17. doi: 10.14745/ccdr.v42is1a03

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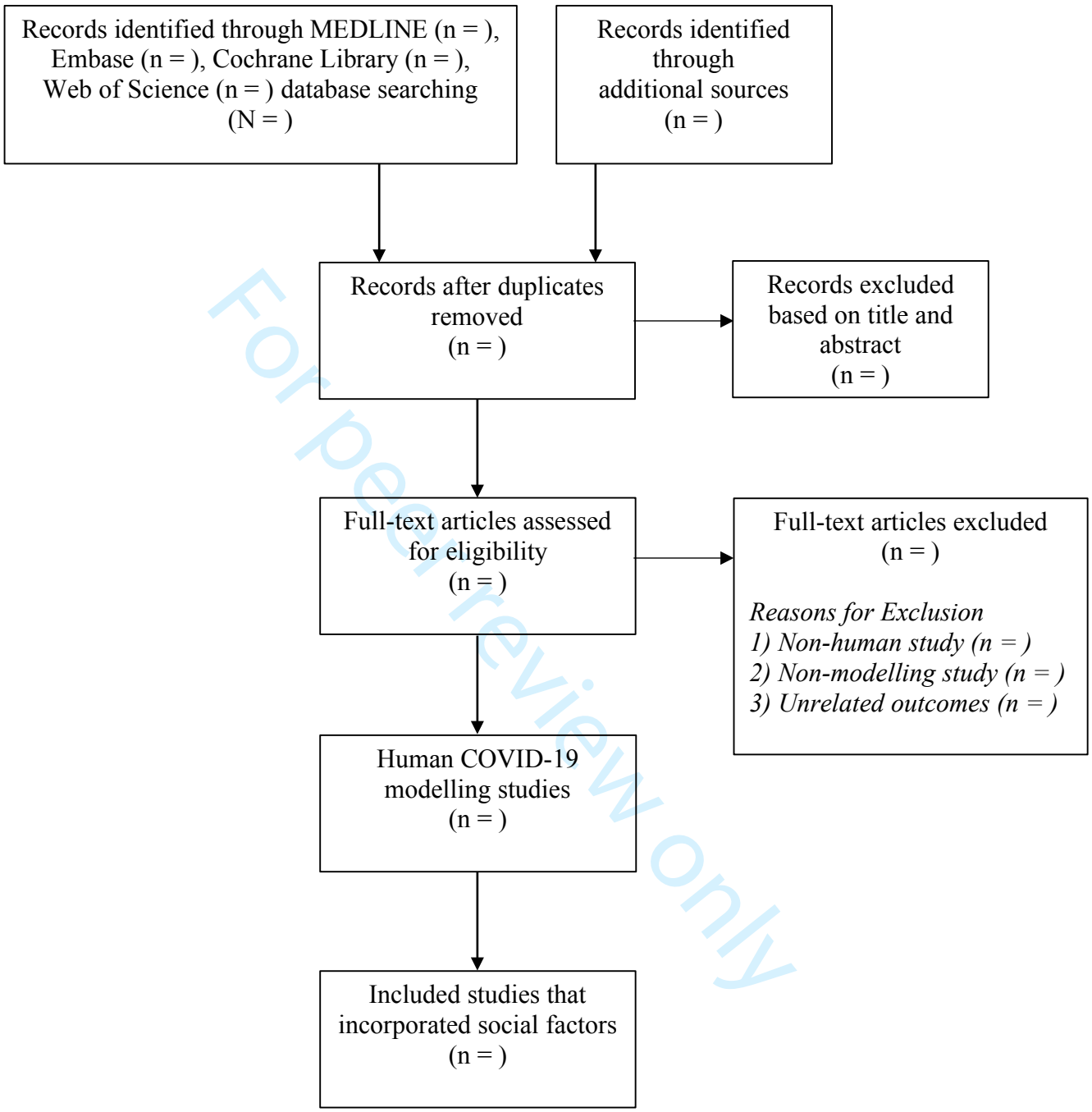


Figure 1. Flow diagram of selection criteria.

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		Reporting Item	Page Number
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Identification	#1a	Identify the report as a protocol of a systematic review	Title page
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	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	4
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Title page
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	11
Amendments			

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	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n/a
Support			
Sources	#5a	Indicate sources of financial or other support for the review	11
Sponsor	#5b	Provide name for the review funder and / or sponsor	11
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	n/a
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	1-3
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	3
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	4
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	4
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	5
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	6
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	6

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Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	6
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	6
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	6
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	8
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	n/a
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	n/a
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	n/a
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	8
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	n/a
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	n/a

None The PRISMA-P checklist is distributed under the terms of the Creative Commons Attribution License CC-BY 4.0. This checklist can be completed online using https://www.goodreports.org/, a tool made by the EQUATOR Network in collaboration with Penelope.ai

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Are COVID-19 models blind to the social determinants of health? A systematic review protocol

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Are COVID-19 models blind to the social determinants of health? A systematic review protocol

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ABSTRACT

Introduction: Infectious disease models are important tools to inform public health policy decisions. These models are primarily based on an average population approach and often ignore the role of social determinants in predicting the course of pandemic and the impact of policy interventions. Ignoring social determinants in models may cause or exacerbate inequalities. This limitation has not been previously explored in the context of the current pandemic, where COVID-19 has been found to disproportionately affect marginalized racial, ethnic, and socioeconomic groups. Therefore, our primary goal is to identify the extent to which COVID-19 models incorporate the social determinants of health in predicting outcomes of the pandemic.

Methods and analysis: We will search MEDLINE, EMBASE, Cochrane Library, and Web of Science databases from December 2019 to August 2020. We will assess all infectious disease modelling studies for inclusion of social factors that meet the following criteria: a) focused on human spread of SARS-CoV-2; b) modelling studies; c) interventional or non-interventional studies; and d) focused on one of the following outcomes: COVID-19 related outcomes (e.g. cases, deaths, etc.), non-COVID-19 related outcomes (i.e. impacts of the pandemic or control policies on other health conditions or health services), or impact of the pandemic or control policies on economic outcomes. Data will only be extracted from models incorporating social factors. We will report the percentage of models that considered social factors, indicate which social factors were considered, and describe how social factors were incorporated into the conceptualization and implementation of the infectious disease models. The extracted data will also be used to create a narrative synthesis of the results.

Ethics and dissemination: Ethical approval is not required as only secondary data will be collected. The results of this systematic review will be disseminated through peer-reviewed publication and conference proceedings.

Systematic review registration: PROSPERO CRD42020207706

Keywords: COVID-19, epidemiology, public health, social medicine

ARTICLE SUMMARY

Strengths and limitations of this study

- The PRISMA-P (Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols) will be followed.
- A systematic approach to evaluating modellers' incorporation of social factors into COVID-19 models will identify practical approaches to better quantify social inequities of the pandemic and predict the impact of public policies on marginalized populations.
- Included studies of COVID-19 models may not be amenable to quantitative synthesis due to heterogeneity of the literature.
- Given the timeframe of the search strategy and the rapid evolution of COVID-19 modelling literature, the results may not include studies evaluating certain prevention strategies, such as vaccination.

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INTRODUCTION

In the early months of the pandemic, COVID-19 was described by New York Governor Andrew Cuomo as the 'great equalizer'[1] among all individuals as the new coronavirus swept through society without bias. A shared lack of immunity meant that regardless of prestige, wealth, fame, or age, we were all equally at-risk.[1] However, epidemiological researchers quickly demonstrated that this was, and remains to be, far from the truth. The pandemic has put existing health inequalities under the microscope and highlighted that COVID-19 has disproportionately affected marginalized racial, ethnic, and socioeconomic groups through its unequal health burden and its disparity of economic losses.[2, 3] A recent investigation found that among 158 counties in large U.S. metropolitan areas, those with predominantly non-White populations had infection rates approximately eight times higher, and a fatality rate more than nine times higher than counties with a mostly White population.[4] Similarly, people living in the poorest neighbourhoods of England and Wales have been found to be twice as likely to die from COVID-19 as those in more affluent areas.[5] Examples of similar trends can be found throughout history. Perhaps most notably in terms of size and scale, data collected from the 1918 flu pandemic demonstrated that non-White minorities had higher all-cause mortality and influenza-related mortality rates when compared to their White counterparts.[6] These disparities are a symptom of deeper societal and health system inequities, including disproportionate exposure through high risk jobs, prevalence of comorbidities, and inequitable access to testing and treatment.

Over the past century, infectious disease models have played a fundamental role in helping public health policy makers implement infection control interventions, as well as shape the theories utilized in disease mitigation.[7] Infectious disease models have been described as,

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"a scientific approach to formulating an explanation for an observed phenomenon and then testing this formulation to project the outcome of various experiments under pertinent conditions.[8(p1)]" Models provide a framework for experts to study a number of important factors related to the SARS-CoV-2 pandemic, including the following:

1) Viral transmission dynamics – models use observed data such as the number of confirmed cases, to estimate difficult to observe transmission dynamics, such as the incubation period and percentage of asymptomatic spread in the population[9, 10]
2) Anticipate the future course of an outbreak – models forecast future incidence, prevalence, morbidity and mortality, based on input parameters[11, 12]
3) Guide public health planning and infectious disease control – models help researchers simulate real-world possibilities and thus appraise the potential impact of disease control strategies such as mask wearing, school closures, testing and contract-tracing on the population[13, 14]
4) Models can estimate a range of outcomes, those directly related to COVID-19 or secondary effects of the disease and control measures, on population health, health service utilization and economic outcomes such as cost and quality of life.[15]

Infectious disease models can only be fit for purpose if the models are appropriately conceptualized and model parameters are based on the best available evidence.[16, 17] Model conceptualization requires decisions about the model structure, including which health states to model, division of the population into sub-groups and selection of characteristics to define the subgroups.[16] Thus, models can incorporate the social determinants of health,[18] such as income, education, employment, and ethnicity. However, structuring models to reflect social determinants increases complexity and sometimes, due to limited availability of data, raises

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questions about structural uncertainty.[19, 20] Complexity and uncertainty may lead modellers to simplify and favour average population models foregoing appraisal of different types of individuals or important subgroups.[19,20,21] Current evidence highlights the importance of social factors in predicting the course of the pandemic and in analyses of policy options,[2-6] although it is unclear to what extent current infectious disease models are based on an average population approach and ignore social factors. To the best of our knowledge, this limitation has yet to be explored in the context of the COVID-19 pandemic. Therefore, the present systematic review will aim to answer the following research question: "To what extent do COVID-19 models incorporate the social determinants of health?" Our objective is to systematically review human COVID-19 modelling (predictive, forecasting) to identify what proportion of models incorporate social factors and for included modelling studies, to identify how social factors were incorporated into the conceptualization of the infectious disease models. The work will provide an important reflection on COVID-19 modelling practices and can provide a blueprint for other modellers to incorporate social factors in the future. We hope that this review will lead to infectious disease models that explicitly acknowledge the role of social determinants in making better predictions as well as mitigating inequalities, and therefore result in better infectious disease outcomes.

METHODS

Protocol and registration

This systematic review will be conducted according to the Preferred Reporting Items for Systematic Reviews and Meta-analyses (PRISMA) guidelines[22] for the identification, screening, eligibility, and inclusion of retrieved research studies. The current protocol is registered in the International Prospective Register of Systematic Reviews (PROSPERO)[23]

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(Registration number CRD42020207706). The anticipated start date for the proposed systematic review is November 2020. The full results will be submitted for peer-reviewed publication immediately following completion of quality assessment and the drafting of a manuscript in May 2021.

Search strategy and eligibility criteria

MEDLINE, EMBASE, Cochrane Library, and Web of Science databases will be searched. The present systematic review is designed to have broad eligibility criteria to identify all COVID-19 modelling studies published from December 2019 to August 14, 2020. Studies will be initially eligible if they: 1) have a human SARS-CoV-2 population, 2) are a modelling study (e.g. mechanistic, mathematical, network, simulation, auto-regressive, Markov, or other model), and 3) investigate one of the following outcomes: COVID-19 disease-related outcomes (e.g. spread, hospitalizations, deaths, recoveries, etc.), non-COVID-19 disease-related outcomes (e.g. impact on other health conditions such as mental health), impacts on health services (e.g. bed occupancy, ventilators, surgical delays), impact of policies (e.g. lock-downs, social distancing) or interventions (vaccines, treatments, etc.) on COVID-19 outcomes or its societal impact. Eligible studies will then be assessed for components of social determinants. This will allow the research team to calculate the percentage of COVID-19 modelling studies that incorporated social determinants into their predicted outcomes of the pandemic (i.e. studies that included social factors (numerator) and all eligible studies (denominator)). Data will only be extracted from models that included social determinants. The following studies will be excluded: 1) within-host biological studies, 2) non-human SARS-CoV-2 studies, 3) phylogenic/genetic studies, 4) environmental/meteorological studies without health impact analysis on humans, 5)

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epidemiological/statistical analyses without a model, and 6) economic analyses without a model.

An example search strategy can be found in Table 1.

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Table 1. MEDLINE search strategy (August 4, 2020)

1. coronavirus/ or betacoronavirus/ or coronavirus infections/ (20740)
2. ("coronavirus*" or "coron?virinae*" or "novel coronavirus*" or "novel corona virus" or "Severe Acute Respiratory Syndrome Coronavirus 2" or "coronavirus disease 2019" or "coronavirus pandemic?" or "coronavirus epidemic?" or "coronavirus outbreak?" or "corona virus pandemic?" or "corona virus epidemic?" or "corona virus outbreak?" or "corona virus disease 2019" or "new coronavirus" or "new corona virus*" or "new coronaviruses" or "novel coronaviruses" or "2019 ncov" or "nCov 2019" or "SARS Coronavirus 2" or "2019-nCoV" or "2019nCoV" or "2019-CoV" or "nCoV2019" or "nCoV-2019" or "COVID-19" or "COVID19" or "CORVID-19" or "CORVID19' or WN-CoV or WNCoV'" or "HCoV-19" or "HCoV19" or "CoV " or "2019 novel*" or "Ncov" or "nCov" or "n-cov" or "SARSCoV-2" or "SARSCoV-2" or "SARSCoV2" or "SARS-CoV2" or "SARS-COV-2" or "SARSCov19" or "SARS-Cov19" or "SARSCov-19" or "SARS-Cov-19" or "SARSr-cov" or "Ncovor" or "Ncorona*" or "Ncorono*" or "NcovWuhan*" or "NcovHubei*" or "NcovChina*" or "NcovChinese*" or "Wuhan virus*" or "novel CoV" or "CoV 2" or "CoV2" or "betacoron?vir*").mp. (54405)
3. ((corona* or corono*) adj1 (virus* or viral* or virinae*)).mp. (1032)
4. (((respiratory* adj2 (acute* or symptom* or disease* or illness* or infect* or condition*)) or "sea-food market*" or "seafood market*" or "food market*" or "foodmarket*" or "wet market*" or "wet-market*" or "wetmarket*") adj10 (Wuhan* or Hubei* or China* or Chinese* or Huanan*)).mp. (1479)
5. ((outbreak* or wildlife* or wild-life or pandemic* or epidemic* or coronavirus or corona virus) adj3 (Wuhan* or Hubei* or China* or Chinese* or Huanan*)).mp. (2228)
6. (anti-flu* or anti-influenza* or antiflu* or antinfluenza*).mp. (3379)
7. or/1-6 (60113) - **(All Coronavirus / COVID-19 Terms)**
8. Models, Theoretical/ (150928)
9. Models, Biological/ (338438)
10. Stochastic Processes/ or stochastic.mp. (44593)
11. (forecast* or model* or simulat*).af. (4238913)
12. or/8-11 (4250384) - **(All Model Terms)**
13. 7 and 12 (6971) - **(All Coronavirus / COVID-19 Terms) AND (All Model Terms)**
14. transmi*.ti,kf. (110754)
15. 7 and 14 (1788) – **(Result Set to capture a non-COVID-19 paper)**
16. 13 or 15 (8442)
17. ("32691014" or "32689711" or "32171948" or "32687538" or "32685697" or "32685143" or "32673577" or "32660125" or "32546824" or "32372755" or "32246905" or "32325039" or "32444481" or "32269020" or "32245814" or "32632012" or "22999128" or "22990082" or "31422772" or "28577700" or "24034486" or "32691015" or "32690354" or "32680824" or "32298421" or "22999128" or "22990082" or "22999132" or "31422772" or "28577700" or "24034486").ui. (26)
18. 13 and 17 (20)
19. limit 16 to yr="2019 -Current" (4850) - **(Final Result Set)**

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Review process

After executing the search strategy, articles will be uploaded to Covidence Software (v204501fd4fa9) and duplicates will be removed. Titles and/or abstracts of the articles will be screened independently by two review members to identify studies that potentially meet the inclusion criteria outlined above. Conflicts will be resolved by a third independent reviewer. The full texts of these potentially eligible studies will then be retrieved and independently assessed for eligibility by two review team members. Any disagreements between them over the eligibility of particular studies will be resolved by a third reviewer. Data will then be extracted from the studies that incorporated social factors. A flow chart for this systematic review can be found in Figure 1.

Data extraction and quality assessment

Data will be extracted into Covidence by one reviewer and confirmed by a second reviewer. Disagreements will be resolved with a discussion between the two reviewers and if necessary, the final decision will be made by a third reviewer. If data related to the outcomes are missing, this will be reported. Authors will not be contacted to request missing data. The following data will be extracted from included studies: author, publication year, geography (country, province/state, city/town, region, low/middle/high income country), setting (Ex. general population, long-term care home, hospital, workplace etc.), population size, goals (predicting the pandemic, impact on other health conditions, impact on health services, impact of policy or interventions, economic analyses), type of model (mathematical, mechanistic, etc.), social and economic factors, and approach to incorporating social and economic factors. An example extraction table containing the characteristics of included studies in the systematic review can be found in Table 2.

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Table 2. Example Extraction Table - Characteristics of Included Studies

Author(s)	Month/ Year	Geography	Setting	(N)	Goal(s)	Model Type	Social Factors	Approach to Social Factors
Smith et al.	March 2020	Milan, Italy High income	Hospital	400	Predicting the Pandemic Case Fatality	Compartmental	Income Employment	TBD
Jones et al.	April 2020	Wuhan, China High income	General	2000	Impact of Policy or Intervention Lock-down	Network	Education	TBD
Harrison et al.	May 2020	Seattle, U.S. High income	Long-term care	300	Impact on other health conditions Mental Health	Mechanistic	Employment	TBD

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There is currently no standardized tool for assessing the quality of infectious disease modelling studies. We plan to adapt the principles of best practices for infectious disease modelling, laid out by Pitman and colleagues into a tool for quality assessment of infectious disease models.[24]

Evidence synthesis

The extracted data will be used to create a narrative synthesis of the results. Based on our broad inclusion criteria, it is likely that the study designs and data will be too heterogeneous for meta-analyses to be appropriate. We expect to report on the number (percent) of identified infectious disease models that incorporated social factors. We will report on the social factors considered, and indicate the number (percent) of included models that considered each social factor. The narrative synthesis will describe how social factors were incorporated into the conceptualization of the infectious disease models, estimate policy impact across social groups (if relevant), and describe which model types (e.g. compartmental, network models) were used when social factors were considered. We will compare models in terms of the approaches used to incorporate social factors; identify trends and see if an increasing number of published models incorporated social factors over time; and based on our knowledge of published epidemiological estimates that quantify the association between social factors and COVID-19,[25] compare incorporation of social factors in COVID-19 models to this evidence base.

DISCUSSION

To the best of our knowledge, this will be the first systematic review that aims to identify the frequency and approach used in COVID-19 models to incorporate the social determinants of health to predict outcomes of the pandemic. Despite previous research and adequate knowledge of the disproportionate health burden experienced by marginalized groups, we hypothesize that

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the majority of COVID-19 models continue to ignore these important social factors. Without taking into account the influences of unequal levels of exposure, vulnerability, and inequitable access to health services dictated by social factors,[26] COVID-19 models may be providing an unclear illustration of the impact of COVID-19 and disease control efforts. When health system decision-makers utilize average population-based models to implement large-scale health policies, they may be unintentionally exacerbating existing health inequities and leaving these marginalized groups further behind. Furthermore, failure to incorporate social factors may lead to missed opportunities to appraise targeted interventions to overcome inequities in the impact of COVID-19. Indeed, interventions targeted to socially vulnerable groups may prove more efficient for controlling the pandemic than interventions at the population level. Although our primary goal is to solidify this fact into scientific literature, our secondary goal with this review is to identify modellers that have incorporated social factors into the conceptualization of their COVID-19 models, and detail the approach they used to do so. It is plausible that the lack of social factors incorporated into COVID-19 models is not a result of neglect, but due to time constraints and limited computational power to handle increased model complexity. The modelling community is highly specialized and their skill sets are coveted by a range of policy decision-makers. Thus, modellers can have significant influence over policy decision-making processes. If modellers increase the awareness of the need to incorporate social factors into COVID-19 models, constraints on incorporating social factors into models can potentially be overcome through increased investments in computer infrastructure and strategic collaboration amongst modelling groups. It is our hope that this review will help provide a preliminary roadmap for modellers to incorporate social factors into future COVID-19 models, and subsequently result in better health outcomes for the entire population.

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The proposed systematic review will likely be limited by the rapidly evolving nature of COVID-19 modelling literature. Given the timeframe of the search strategy, we will likely not find many studies evaluating vaccination strategies. However, the findings on incorporating social factors into infectious disease models will still be relevant to modellers working on vaccine policy appraisal.

The strengths of the proposed review are that a systematic approach to evaluating modellers' incorporation of social factors into COVID-19 models can identify practical approaches to making better predictions and reduce inequities, quantify gaps in the literature, motivate enhancements in modelling best practices, and ultimately improve health for socially and economically vulnerable groups related to COVID-19 and other health conditions.

ETHICS AND DISSEMINATION

Ethical approval is not required as only secondary data will be collected. The results of this systematic review will be disseminated through peer-reviewed publication and conference proceedings.

PATIENT AND PUBLIC INVOLVEMENT

There were no time or funds allocated to patient and public involvement due to safety concerns in the context of the current COVID-19 pandemic. However, this systematic review asks an important research question relevant to the health and protection of vulnerable groups.

Author Contributions: AJB and SA secured the funding, designed the review, drafted the paper, and approved the final version to be published. MSM wrote the paper and approved the final version to be published.

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Competing Interests: None declared.

Figure Captions:

Figure 1. Flow diagram of selection criteria.

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REFERENCES

1. Mein SA. COVID-19 and Health Disparities: the Reality of "the Great Equalizer". *J Gen Intern Med* 2020;35(8):2439-40. doi: 10.1007/s11606-020-05880-5

2. Ali S, Asaria M, Stranges S. COVID-19 and inequality: are we all in this together?. *Can J Public Health* 2020;111:415-416. doi: 10.17269/s41997-020-00351-0

3. Williams DR, Cooper LA. COVID-19 and health equity—a new kind of "herd immunity". *JAMA* 2020;323(24):2478-2480. doi: 10.1001/jama.2020.8051

4. Adhikari S, Pantaleo NP, Feldman JM, et al. Assessment of community-level disparities in coronavirus disease 2019 (COVID-19) infections and deaths in large US metropolitan areas. *JAMA Network Open* 2020;3(7):e2016938-e2016938. doi: 10.1001/jamanetworkopen.2020.16938

5. Caul S. Deaths involving COVID-19 by local area and socioeconomic deprivation: deaths occurring between 1 March and 31 July 2020. Statistical Bulletin. 2020.

6. Hutchins SS, Fiscella K, Levine RS, et al. Ompad, D. C., & McDonald, M. Protection of racial/ethnic minority populations during an influenza pandemic. *Am J Public Health* 2009;99(S2):S261-S270. doi: 10.2105/AJPH.2009.161505

7. Lessler J, Cummings DA. Mechanistic models of infectious disease and their impact on public health. *Am J Epidemiol* 2016;183(5):415-422. doi: 10.1093/aje/kww021

8. Star L, Moghadas SM. The role of mathematical modelling in public health planning and decision making. Purple Paper, National Collaborative Center for Infectious Diseases. 2010.

9. Zhou L, Liu JM, Dong XP, McGoogan JM, et al. & Wu, Z. Y. COVID-19 seeding time and doubling time model: an early epidemic risk assessment tool. *Infect Dis Poverty* 2020; 9(76):1-9. doi: 10.1186/s40249-020-00685-4

10. Tuite AR, Fisman DN. Reporting, epidemic growth, and reproduction numbers for the 2019 novel coronavirus (2019-nCoV) epidemic. *Ann Intern Med* 2020;172(8):567-568. doi: 10.7326/M20-0358

11. Wang Y, Xu C, Yao S, et al. Forecasting the epidemiological trends of COVID-19 prevalence and mortality using the advanced α-Sutte Indicator. *Epidemiol Infect* 2020;148:e236. doi: 10.1017/S095026882000237X

12. Ogden NH, Fazil A, Arino J, et al. Artificial intelligence in public health: Modelling scenarios of the epidemic of COVID-19 in Canada. *Can Commun Dis Rep* 2020;46(8):198-204. doi: 10.14745/ccdr.v46i06a08

13. Fisman DN, Greer AL, Tuite AR. Bidirectional impact of imperfect mask use on reproduction number of COVID-19: A next generation matrix approach. *Infect Dis Model* 2020;5:405-408. doi: 10.1016/j.idm.2020.06.004

14. Tuite AR, Fisman DN, Greer AL. Mathematical modelling of COVID-19 transmission and mitigation strategies in the population of Ontario, Canada. *CMAJ* 2020;192(19): E497-E505. doi: 10.1503/cmaj.200476

15. Wells CR, Fitzpatrick MC, Sah P, et al. Projecting the demand for ventilators at the peak of the COVID-19 outbreak in the USA. *Lancet Infect Dis* 2020;20(10):1123-1125. doi: 10.1016/S1473-3099(20)30315-7

16. Roberts M, Russell LB, Paltiel AD, et al. Conceptualizing a model: a report of the ISPOR-SMDM modeling good research practices task force–2. *Med Decis Making* 2012;32(5): 678-689. doi: 10.1177/0272989X12454941

17. Briggs AH, Weinstein MC, Fenwick EA, et al. Model parameter estimation and uncertainty analysis: a report of the ISPOR-SMDM Modeling Good Research Practices Task Force Working Group–6. *Med Decis Making*, 2012;32(5):722-732. doi: 10.1177/0272989X12458348

18. Marmot M. Social determinants of health inequalities. *Lancet Public Health* 2005;365(9464): 1099-1104. doi: 10.1016/S0140-6736(05)71146-6

19. Mauskopf J. Multivariable and structural uncertainty analyses for cost-effectiveness estimates: back to the future. *Value Health* 2019;*22*(5): 570-574. doi: 10.1016/j.jval.2018.11.013

20. Jain R, Grabner M, Onukwugha E. Sensitivity analysis in cost-effectiveness studies. *Pharmacoeconomics* 2011;29(4):297-314. doi: 10.2165/11584630-000000000-00000

21. Sculpher M. Subgroups and heterogeneity in cost-effectiveness analysis. *Pharmacoeconomics* 2008;26(9):799-806. doi: 10.2165/00019053-200826090-00009

22. Mohers D, Shamseer L, Clarke M, et al. Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement. *Syst Rev* 2015;*4*(1): 1. doi: 10.1186/2046-4053-4-1

23. Booth A, Clarke M, Dooley G, et al. The nuts and bolts of PROSPERO: an international prospective register of systematic reviews. *Syst Rev* 2012;1(2): 1-9. doi: 10.1186/2046-4053-1-2

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24. Pitman R, Fisman D, Zaric GS, et al. Dynamic transmission modeling: a report of the ISPOR-SMDM Modeling Good Research Practices Task Force Working Group–5. *Med Decis Making* 2012;32(5):712-721. doi: 10.1177/0272989X12454578

25. Upshaw TL, Brown C, Smith R, et al. Social determinants of COVID-19 incidence and outcomes: A rapid review. *PLOS One*. 2021;16(3):e0248336. doi: 10.1371/journal.pone.0248336

26. Moloughney BW. Social Determinants of Health: What can public health do to address inequities in infectious disease?. *Can Commun Dis Rep* 2016;42(Suppl 1): S1-14-S1-17. doi: 10.14745/ccdr.v42is1a03

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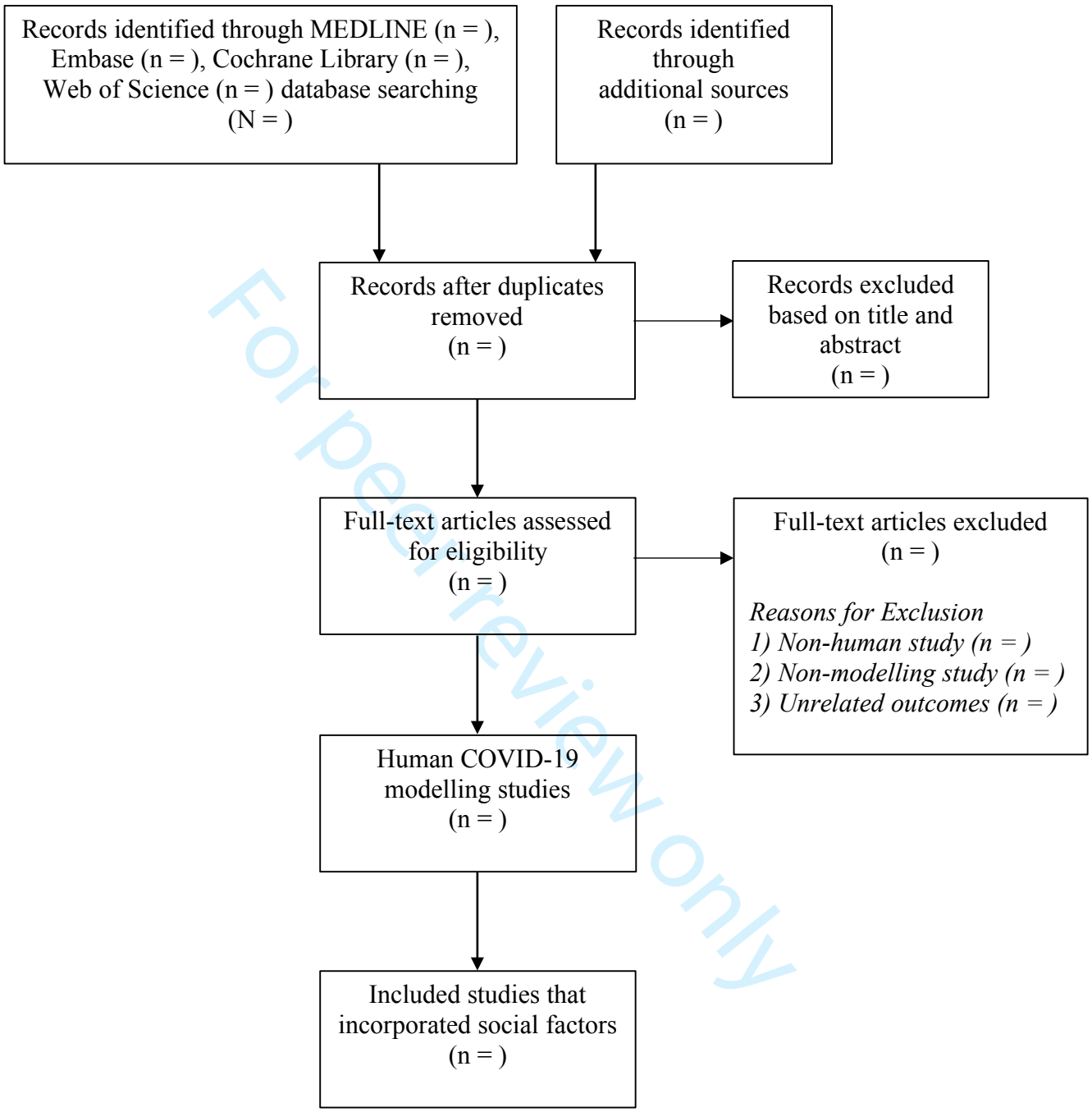


Figure 1. Flow diagram of selection criteria.

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Reporting checklist for protocol of a systematic review.

Based on the PRISMA-P guidelines.

Instructions to authors

Complete this checklist by entering the page numbers from your manuscript where readers will find each of the items listed below.

Your article may not currently address all the items on the checklist. Please modify your text to include the missing information. If you are certain that an item does not apply, please write "n/a" and provide a short explanation.

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In your methods section, say that you used the PRISMA-Preporting guidelines, and cite them as:

Moher D, Shamseer L, Clarke M, Ghersi D, Liberati A, Petticrew M, Shekelle P, Stewart LA. Preferred Reporting Items for Systematic Review and Meta-Analysis Protocols (PRISMA-P) 2015 statement. Syst Rev. 2015;4(1):1.

		Reporting Item	Page Number
Title			
Identification	#1a	Identify the report as a protocol of a systematic review	Title page
Update	#1b	If the protocol is for an update of a previous systematic review, identify as such	n/a
Registration			
	#2	If registered, provide the name of the registry (such as PROSPERO) and registration number	4
Authors			
Contact	#3a	Provide name, institutional affiliation, e-mail address of all protocol authors; provide physical mailing address of corresponding author	Title page
Contribution	#3b	Describe contributions of protocol authors and identify the guarantor of the review	10
Amendments			

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	#4	If the protocol represents an amendment of a previously completed or published protocol, identify as such and list changes; otherwise, state plan for documenting important protocol amendments	n/a
Support			
Sources	#5a	Indicate sources of financial or other support for the review	11
Sponsor	#5b	Provide name for the review funder and / or sponsor	11
Role of sponsor or funder	#5c	Describe roles of funder(s), sponsor(s), and / or institution(s), if any, in developing the protocol	n/a
Introduction			
Rationale	#6	Describe the rationale for the review in the context of what is already known	1-3
Objectives	#7	Provide an explicit statement of the question(s) the review will address with reference to participants, interventions, comparators, and outcomes (PICO)	3
Methods			
Eligibility criteria	#8	Specify the study characteristics (such as PICO, study design, setting, time frame) and report characteristics (such as years considered, language, publication status) to be used as criteria for eligibility for the review	4
Information sources	#9	Describe all intended information sources (such as electronic databases, contact with study authors, trial registers or other grey literature sources) with planned dates of coverage	4
Search strategy	#10	Present draft of search strategy to be used for at least one electronic database, including planned limits, such that it could be repeated	6
Study records - data management	#11a	Describe the mechanism(s) that will be used to manage records and data throughout the review	7
Study records - selection process	#11b	State the process that will be used for selecting studies (such as two independent reviewers) through each phase of the review (that is, screening, eligibility and inclusion in meta-analysis)	7

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Study records - data collection process	#11c	Describe planned method of extracting data from reports (such as piloting forms, done independently, in duplicate), any processes for obtaining and confirming data from investigators	7
Data items	#12	List and define all variables for which data will be sought (such as PICO items, funding sources), any pre-planned data assumptions and simplifications	7
Outcomes and prioritization	#13	List and define all outcomes for which data will be sought, including prioritization of main and additional outcomes, with rationale	7
Risk of bias in individual studies	#14	Describe anticipated methods for assessing risk of bias of individual studies, including whether this will be done at the outcome or study level, or both; state how this information will be used in data synthesis	7
Data synthesis	#15a	Describe criteria under which study data will be quantitatively synthesised	n/a
Data synthesis	#15b	If data are appropriate for quantitative synthesis, describe planned summary measures, methods of handling data and methods of combining data from studies, including any planned exploration of consistency (such as I2, Kendall's τ)	n/a
Data synthesis	#15c	Describe any proposed additional analyses (such as sensitivity or subgroup analyses, meta-regression)	n/a
Data synthesis	#15d	If quantitative synthesis is not appropriate, describe the type of summary planned	8
Meta-bias(es)	#16	Specify any planned assessment of meta-bias(es) (such as publication bias across studies, selective reporting within studies)	n/a
Confidence in cumulative evidence	#17	Describe how the strength of the body of evidence will be assessed (such as GRADE)	n/a

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